

INTELLECTUAL PROPERTY RIGHTS: PATENTS AND PATENTING

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ABSTRACT

Intellectual Property Rights (IPR) is a territorial rights offered by the government to an inventor for the protection of its intellectual ideas and creativity. In simple words IPR can be formulated as; Intellectual Creation + Protection = Intellectual Property Rights. There are five basic types of intellectual property rights via patents, copyrights, trademarks, geographical indicators and (NPV) new plant varieties. This paper mainly focuses on patents and its significance from the Indian patenting perspective. We have discussed three case studies in support of the patenting significance and explained a couple of examples to better understand the definition of patent.

Key words: Patents, Patentability, Indian Patent System, Innovators statistics

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1. INTRODUCTION

Intellectual property (IP) is a legal right that allows people to own their innovation and creativity in the similar way that one can own a physical property like lands, shops, flats, etc. The IP owner can be often rewarded for the innovation which encourages the owner for the further innovation and creativity to benefit the nation. In general IP gives rise to protection for ideas but, it will have to be more elaboration of an idea before protection can arise. It will not be easily possible to protect IP and gain IP rights unless they have been applied for and granted by patent office.

Till recently Intellectual property Rights was categorized broadly into patents, copyrights, industrial designs and trade secrets, covered under various multilateral agreements. But in the year 1994 TRIPs agreement brought out a comprehensive listing of IPRs. These improvements are expected to influence competition in the global economy and catalyse generations and diffusion of technological innovations for the benefit of the country. The IPRs can be divided into two major categories:

1.1. Industrial Property

Which includes Inventions (patents), Integrated Circuit Layout Design, Trademarks, Plant varieties, Industrial Designs, and Geographic Indications of source,

1.2. Copyright

Which includes artistic and literary such as films, novels, poems and plays, musical works, artistic works such as drawings, paintings, photographs and sculptures, architectural designs and Scientific works and theories. Related rights to copyright include those of performing artists in their performances, producers of phonograms in their recordings, broadcasters in their radio and television programs, etc.

1.3. Example

Consider an example of the electronic devices as shown in table 1 to better understand the intellectual property rights (IPR)

Table 1 Showing IPRs for the electronic devices

Sr. No	Intellectual Property	Electronic Devices	
		Smartphone	Laptop
1.	Copyright	Operating systems (Java, Android, etc) User Interface.	Operating systems (Linux, Windows, etc) User Interface.
2.	Design	Outer shape	Outer Shape, Cabinet
3.	Trademark	Logo, Brand names (Apple, Nokia, etc)	Logo, Brand names (Dell, HP, etc)
4.	Patent	Hardware circuit board, processor, integrated circuits(SMDs, chips), etc	Hardware circuit board, processor, integrated circuits(SMDs, chips), etc

2. PATENTS

Patent is a statutory right granted by the government to the inventor for an invention, which is a product or a process, in general a new way of doing something or it offers a new technical solution to a problem. This right is granted by the countries' or regional patent office in exchange of complete disclosure of the invention. A patent provides protection for the invention to the owner of the patent means that the invention cannot be commercially made, sold, used or distributed or without the patent owner's consent. These patent rights are usually enforced in a court, which holds the authority to stop patent infringement. Conversely, a court can also declare a patent invalid upon a successful challenge by a third party. The patent protection is granted for a limited period, generally twenty years.

2.1. Patent Owner's Rights

- The patent owner is not given statutory rights to practice his invention, but has rights to prevent others from commercially exploiting his invention. Basically patent owner has complete rights to exclude others from making, using or selling the invention.
- Patent owner has the authority to decide who may - or may not – use the patented invention for the period in which the invention is protected.
- The patent owner may give license, to the parties to use the invention on mutually agreed terms.

- The patent owner has a right to sell the invention to someone else, who will then become the new owner of the patent.
- Once a patent expires, the protection ends, all the rights of the patent owner are expired and an invention enters into the public domain, which becomes available for commercial exploitation by others.

2.2. Need of Patents

Patents provides incentives to the inventor by the means of recognition for their creativity and materialistic rewards for their invention which has a potentially of commercialization. This incentive motivates innovators and encourages them to make more innovations, which continuously enhances the quality of human life.

2.3. Patentability

Any innovation in general must satisfy the condition of novelty, inventive step and practical usage as described below

Novelty: Any innovation must show some new characteristic which is unknown in the body of existing knowledge also called as “prior art”, in its technical field.

Inventive Step: Any innovation must show an inventive step, which could not be deduced by a person with average knowledge of the technical field and also, not obvious to the person skilled in that art, non-anticipated.

Practical Usage: The subject matter of that innovation must be acceptable as “Patentable” under The Patents Act, 1970 with The Patent Rules, 2003 as amended by (Amendment) Rules, 2016.

3. HISTORY OF INDIAN PATENT ACT

Indian patent act has been initiated in India since the year 1856 and depending upon the protection requirements of the innovators, it has been updated for several years as listed in the table 2.

Table 2 Showing year wise history of the Indian patent act.

Sr. No	Year	Patent Act
1.	1856	The Act VI of 1856 on protection of inventions based on the British Patent Law of 1852. Certain exclusive privileges granted to inventors of new manufacturers for a period of 14 years.
2.	1859	The Act modified as act XV Patent monopolies called exclusive privileges (making. Selling and using inventions in india and authorizing others to do so for 14 years from date of filing specification).
3.	1872	The <i>Patents & Designs</i> Protection Act.
4.	1883	The Protection of <i>Inventions</i> Act.
5.	1888	Consolidated as the <i>Inventions & Designs</i> Act.
6.	1911	The <i>Indian Patents & Designs</i> Act.
7.	1972	The Patents Act (Act 39 Of 1970) Came Into Force On 20th April 1972.
8.	1999	On March 26, 1999 <i>Patents (Amendment) Act, (1999)</i> Came Into Force From 01-Jan-1995.
9.	2002	The <i>Patents (Amendment) Act 2002</i> Came Into Force From 20th May 2003
10.	2005	The <i>Patents (Amendment) Act 2005</i> Came Into Force From 4th April 2005

4. IMPORTANT CONVENTIONS/AGREEMENT RELATED TO IPR

Table 3 lists the initiating conventions/agreements/treaties/application types on the specific components and have been followed up by periodic amendments as per needs

Table 3 Showing Conventions related to IPRs.

Sr. No	Name	Year	Subject Matter for Protection
1.	Paris Convention	1883	Industrial Property
2.	Berne Convention	1886	Copyrights of Literary & Artistic Properties
3.	Madrid Agreement	1891	Trademark and Service Marks
4.	Hague Agreement	1925	Industrial Designs
5.	Rome Convention	1961	Performance, Phonograms & Broadcasts
6.	UPOV Convention	1961	Plant Variety
7.	Budapest Treaty	1973	Micro Organisms
8.	Brussels Convention	1974	Satellite Transmitted Programmes
9.	Patent Cooperation Treaty	1978	Patents
10.	Washington Treaty	1989	Integrated Circuits
11.	Trips Agreement	1994	All IPRs and WTO

5. CASE STUDY

CASE STUDY I: Invention of Telescope

Galileo an Italian astronomer, physicist and engineer, sometimes described as a polymath, from Pisa has invented the telescope. A telescope is an optical instrument that makes distant objects appear magnified by using an arrangement of lenses or curved mirrors and lenses, or various devices used to observe distant objects by their emission, absorption, or reflection of electromagnetic radiation. But this instrument was not granted any patent because the claims of this invention were similar to the patents of the spectacles invention. Although another inventor Hans Lippershey has tried to file the patent but was rejected. Reginald Henry Bradshaw invented glasses, monacles, and spectacles and has already filed the patent having the patent number as 1337818, issued in April 1920 for rimless spectacles and eyeglasses. And this filed patent has given rights to Reginald Henry Bradshaw to stop other inventors like Galileo to get the patent for telescope because telescope uses the same concepts of spectacles. So, the conclusion of this case study is that Bradshaw has used the patenting and stopped others from commercially using his intellectuality and creativity.

CASE STUDY II: Invention of Telephone

Telephone was initially named as Teletrofono by Antonio Santi Giuseppe Meucci who was an Italian inventor and an associate of Giuseppe Garibaldi, a major political figure in the history of Italy. Meucci is best known for developing a voice communication apparatus that several sources credit as the first teletrofona. A teletrofono is a telecommunications device that permits two or more users to conduct a conversation when they are too far apart to be heard directly. A telephone converts sound, typically and most efficiently the human voice, into electronic signals that are transmitted via cables and other communication channels to another telephone which reproduces the sound to the receiving user. Meucci first invented teletrofono in 1860 and provisionally filed the Patent after eleven years of invention. And after that he was sick and died before filing the final patent. After two years of his death Alexander

Graham Bell a Scottish-born American inventor, scientist, and engineer has taken up the invention and after making some minor changes he filed the patent. And till now he is credited with inventing and patenting the first practical telegraph renamed as Telephone. So, the conclusion from this case study is that Meucci filed the provisional patent only and lost all the rights of the patent owner, whereas Bell filed the complete and final patent and gain all the rights and benefits of the patent with minor efforts.

CASE STUDY III: Invention of Light Bulb

Light Bulb was originally invented by Nikola Tesla a Serbian-American inventor, electrical engineer, mechanical engineer, and futurist who is best known for his contributions to the design of the modern alternating current (AC) electricity supply system. Light bulb, an incandescent lamp or incandescent light globe is an electric light with a wire filament heated until it glows. The filament is enclosed in a bulb to protect the filament from oxidation. Current is supplied to the filament by terminals or wires embedded in the glass. A bulb socket provides mechanical support and electrical connections. Tesla was not able to patent the invention of light bulb because this invention is similar to his previously patented invention of Arc Lightning systems. And he had already sold his Patent of Arc Lightning system to the Edison Company in exchange of stock. So, Tesla has lost all the rights of patent owner and he was unable to file a patent of addition because he had already sold the invention. After few years Edison Company modified the invention and Patented as Light Bulb on the name of Thomas Alva Edison an American inventor and businessman who has been described as America's greatest inventor. And till now he is being credited for the invention of light bulb. So, the conclusion from this case study is that with the poor knowledge of the patent rights an inventor may lose the invention and other can be the beneficiary without any major efforts.

6. BEST INDIAN TURMERIC TECHNOLOGY

Thiruvananthapuram: Turmeric-based technology to kill cancer cells gets US patent

A potentially breakthrough cancer-fighting technology involving a molecule extracted from turmeric has won Thiruvananthapuram's Sree Chitra Tirunal Institute for Medical Sciences a US patent. According to Lissy Krishnan, head of Sree Chitra's research team, delivery of "curcumin" directly to the affected tissues rather than through conventional oral or intravenous methods enables it to target malignant cancer cells while sparing the healthy ones around them. Turmeric has proven anticancer properties and curcumin, a molecule extracted from it, is easily absorbed by the body and aids blood clotting, Krishnan said. At Sree Chitra, research funded by the Indian Council for Medical Research focused on processing curcumin to form an easy-to-use wafer configuration. When applied to body tissues, the curcumin present in the wafer is released into tissue fluids.

Human albumin, or rich proteins present in the fibrin clot produced from heavy bleeding, binds albumin receptors to cancer cells, thereby permitting its entry into the cells. Simultaneously, the fibrin clot is removed by the body's natural clot breakdown mechanism without any adverse effect. "The fibrin wafer is targeted for implantation into the affected site after surgical removal of cancer tissue for killing any remaining malignant cells that have the potential to cause recurrence of cancer or spread to other parts of the body. In addition to drug delivery, the fibrin wafer can promote blood clotting at the surgical site," said Asha Kishore, director of Sree Chitra.

The institute is ready to transfer the technology for future development of curcumin as an anti-cancer treatment through animal and clinical trials. "The US patent adds value to our

efforts to transfer technology and boosts the industry's confidence in exploring validation and trials with international markets in mind," Kishore said.

7. INDIAN PATENT FILING FLOW

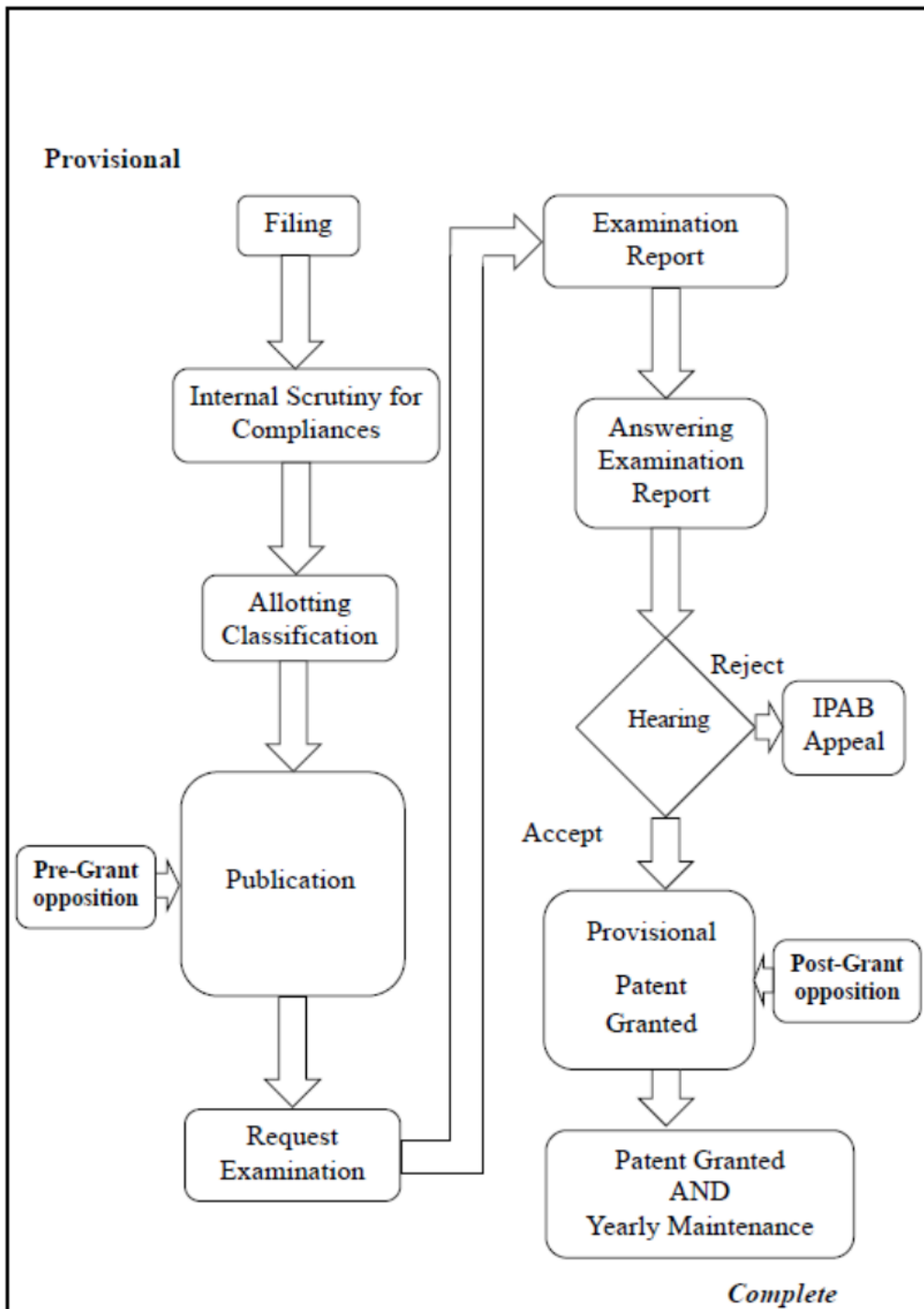


Figure 1 Generalized Block Diagram of the Indian Patent System.

Figure 1 illustrates a generalized block diagram of the Indian Patent system which is explained below.

- The true and first inventor(s), his assignee, or their legal representative – can apply for patent with a provisional or a complete specification either themselves or through an agent at the appropriate office. In India, there are four offices, namely Mumbai, Delhi, Kolkata and Chennai depending upon the address of the applicant where he resides, has a place of business, has his domicile or the place from where invention actually originated.
- Then the patent office starts with the Internal Scrutiny for Compliances with respect to the completion of the application and its supporting documents. And on the basis of the application patent office determines the allocation of the deputy controller and supporting officers to further process the patent application.
- Patent office then publishes the complete patent specification ordinarily after eighteen months from the priority date. Usually it is published on every Friday on the Indian patent website. Optionally a request for early publication can be made whereby publication happens ordinarily within one month of such request.
- Then the application goes for the Examination stage by the patent office. Unlike publication, examination of the patent application is not automatic a request has to be filed by the inventor or its assignee within forty eight months from earlier of date of priority and filing of the patent application.
- Once request for examination has been made patent office examines the claims in support of the oppositions and then, patent office issues examination report to the applicant. Examination of the application is done based on the date of filing of the request for the examination, meaning earlier the request made for the examination, earlier is the report issued.
- After the issuance of first examination report, application is to be put in conditional allowance within twelve months by waiving the objection and complying with requirements (if any).
- After considering replies to the first or subsequent examination reports, post grant oppositions and hearing notices (if any), controller grants or rejects the patent application.
- In India, patent application can be opposed before and after, grant of the patent. Patent can also be revoked under some circumstances even after the grant of the patent.
- Once the patent is granted the inventors gets the monopoly of twenty years on the payment of the necessary fees. Also, renewal fees have to be paid after a twenty years of span in order to keep the patent in force, else it will become open to public if renewal fees are not paid.
- Always an appeal can be made against the decision of the patent office/controller at the Intellectual Property Appellate Board (IPAB) by the aggrieved party in case the grant of the patent application is rejected.

8. PRECAUTIONS PRIOR TO PATENTING

8.1. Never perform any commercial activity on the invention there are possibilities that it can cause infringement

- Never disclose or publish the invention in any of the sources this may cause to kill the creativity of the invention
- Never discuss or share the invention with any of the unauthorized persons this may harm the novelty of the invention
- Always maintain records, results, data which can be easily accessed at the time of opposition done at the patent office. This can be helpful to prove the intellectual of an inventor.
- Always sign a Non-Disclosure Agreement with the person where mandatory disclosure of the invention is needed for the purpose of granting patent.

9. PATENTS AND INNOVATORS STATISTICS

Figure 2 shows the plot of top ten patent filing companies versus the number of patents filed by each company. It can be visualized from the plot that Hitachi Limited has filed the highest number of patent in the year 2017 estimating as Three lakh forty two thousand eight hundred and fifty four.

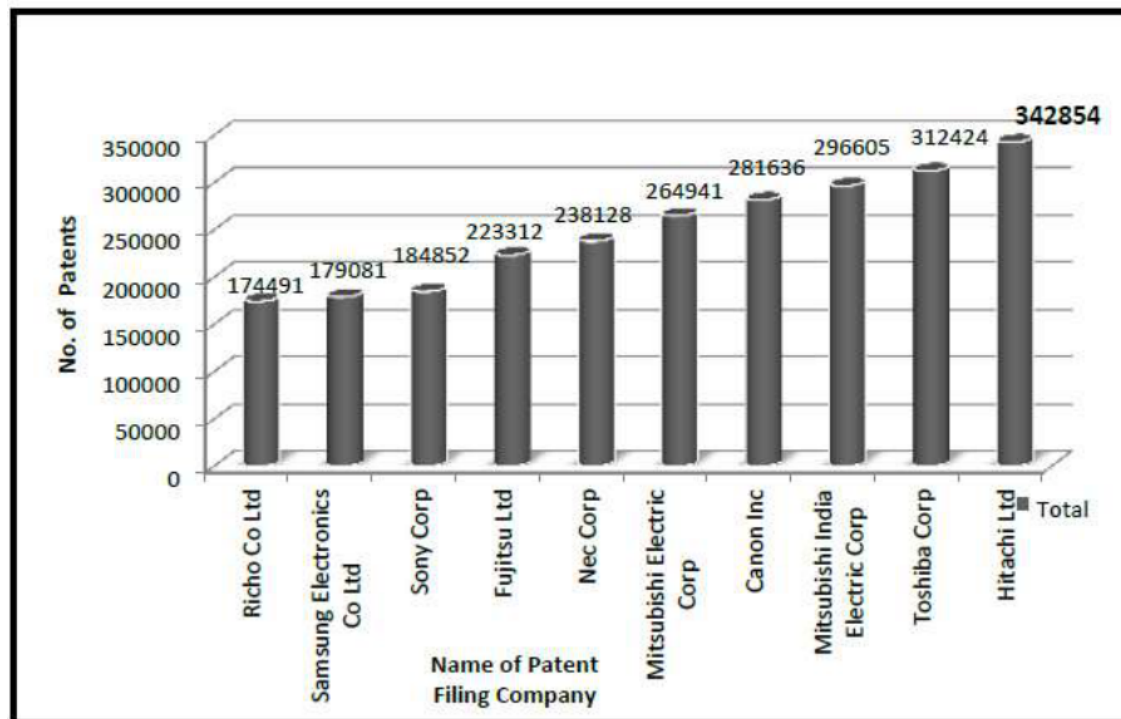


Figure 1 Top TEN patent filing companies

Figure 3 shows the graph of top ten patent filing Indian companies versus the number of patents filed by each company. It can be observed from the plot that Qualcomm Incorporation has filed the highest number of patent in the year 2017 estimating as Nine thousand three hundred and thirty eight.

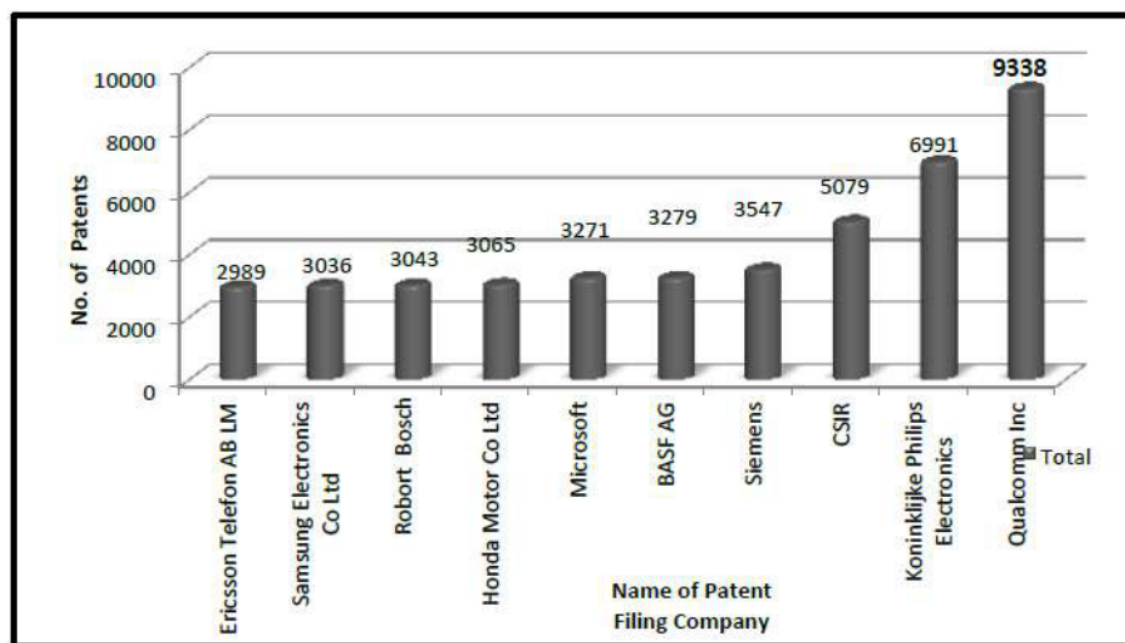


Figure 2 Top TEN patent filing Indian companies

Table 4 shows the record of the country wise top inventors and the count of their patents filed in their country. It can be seen in the table that Kia Silverbrook from Australia has filed the highest number of patents counting as four thousand six hundred and sixty nine and has generated an income of more than One and a half billion dollars from patents in the year 2017.

Table 4 Showing Globally Top Inventors

Sr. No	Name of Inventor	Country	Number of patents
1.	Kia Silverbrook (Net Worth \$1,599,310,958)	Australia	4669
2.	Shunpei Yamazaki	Japan	3516
3.	Paul Lapstun	Australia	1268
4.	Gurtej Sandhu	India	1244
5.	Thomas Edison	USA	1084
6.	Leonard Forbes	Canada	1055
7.	George Albert Lyon	Canada	993
8.	Donald Weder	USA	989
9.	Jun Koyama	Japan	952
10.	John F. O'Connor	USA	949

Table 5 shows the record of the top inventors in INDAI and the count of their patents filed for specific technology in India with respect to total number of the patent filed for that technology in India. It can be observed from the table that Gurtej Sandhu filed the highest number of patents counting as one thousand two hundred and forty four for the thin film processes and materials, VLSI and Semiconductor device fabrication technology in the year range of 1991-2016.

Table 5 INDIAN Top Inventors

Sr. No	Name	No. of Patent	Total Patents	Year Range	Technology
1.	Gurtej Sandhu	1244	2038	1991-2016	Thin film processes and materials, VLSI, Semiconductor device fabrication
2.	Ravi K. Arimilli	506	767	1992-2016	Computer architecture, Semiconductor memory, Cache coherence, Symmetric multiprocessing
3.	Devendra K. Sadana	332	794	1983-2016	Solar cells, OLED, Integrated Circuits, CMOS, DRAM, LEDs

10. SUMMARY

In this paper we have briefly introduced the topic of intellectual property rights (IPR) with more focus on PATENT type of IPR. We have described some real life existing cases for the Invention of telescope, telephone and light bulb which properly explains the significance of patents. In support of the Patenting when have described the flow graph of Indian patenting system. Also, we have illustrated some literature based information of patents and innovators statistically, in support of the topic IPR.

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